

7.1

Marek Adamčík

Zo zadania je jordanova matica rbrane

$$J = \begin{pmatrix} J_1 & 0 & 0 \\ 0 & J_2 & 0 \\ 0 & 0 & J_3 \end{pmatrix}, \text{ kde } J_1, J_2, J_3 \text{ m\u00e1 bloky pr\u00edslu\u0161n\u00fdch velho\u0161t\u00ed}$$

$$J_1 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad J_2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad J_3 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

m\u00e1me b\u00e1zu $B = (b_1, b_2, \dots, b_{10})$

J_1 odpoved\u00e1 vektorom $b_1 \dots b_5$ J_2 vektorom $b_6 \dots b_8$ $J_3 \dots b_9, b_{10}$

Pre jordanove bloky velho\u0161ti m plat\u00ed

$$\dim(\ker f^i) = \min(i, m)$$

pre n\u00e1\u0161 pr\u00edpad teda

$$\dim(\ker f^1) = \min(1, 5) + \min(1, 3) + \min(1, 2) = 1 + 1 + 1 = 3$$

generovan\u00e9 vektormi b_1, b_6, b_9

$$\dim(\ker f^2) = \min(2, 5) + \min(2, 3) + \min(2, 2) = 2 + 2 + 2 = 6$$

generovan\u00e9 $b_1, b_2, b_6, b_7, b_9, b_{10}$

$$\dim(\ker f^3) = \min(3,5) + \min(3,3) + \min(3,2) = 3+3+2 = 8$$

generované $b_1, b_2, b_3, b_6, b_7, b_8, b_9, b_{10}$

$$\dim(\ker f^4) = \min(4,5) + \min(4,3) + \min(4,2) = 4+3+2 = 9$$

... $b_1, b_2, b_3, b_4, b_6, b_7, b_8, b_9, b_{10}$

$$\dim(\ker f^5) = \min(5,5) + \min(5,3) + \min(5,2) = 5+3+2 = 10$$

generované $b_1, \dots, b_{10}$

$$\text{pre } i \geq 5, \dim(\ker f^i) = 10$$

$$\text{ďalej poznáme } \dim(\operatorname{Im} f^j) = \max(m-j, 0), \text{ teda}$$

$$\dim(\operatorname{Im} f^1) = \max(4,0) + \max(2,0) + \max(1,0) = 4+2+1 = 7$$

generované $b_1, b_2, b_3, b_4, b_6, b_7, b_8$

$$\dim(\operatorname{Im} f^2) = \max(3,0) + \max(1,0) + \max(0,0) = 3+1 = 4$$

generované b_1, b_2, b_3, b_6

$$\dim(\operatorname{Im} f^3) = \max(2,0) + \max(0,0) + \max(-1,0) = 2$$

generované b_1, b_2

$$\dim(\operatorname{Im} f^4) = \max(1,0) + \max(-1,0) + \max(-2,0) = 1$$

generované b_1

$$\dim(\operatorname{Im} f^5) = \max(0,0) + \dots = 0$$

$$\text{pre } j \geq 5, \dim(\operatorname{Im} f^j) = 0$$

Teraz môžeme odpozorovať, ktoré vektory máme v priemiku.

- pre $i=1, j=1$

$(\ker f^1) \cap (\operatorname{Im} f^1)$ priemikhoria vektory b_1, b_6, b_9

ktoré sú teda aj bárou tohto priestoru $\dim=3$

- $i=2, j=1$

$(\ker f^2) \cap (\operatorname{Im} f^1)$ priemik b_1, b_2, b_6, b_7, b_9

ktoré teda horia aj bárou tohto priestoru $\dim=5$

Všeobecne

$$\dim((\ker f^i) \cap (\operatorname{Im} f^j)) = \min(i, \max((5-j), 0)) + \min(i, \max((3-j), 0)) + \min(i, \max((2-j), 0))$$

a bárou tohto priemiku bude horiť prvých

- $\min(i, \max((5-j), 0))$ vektorov z bunky J_1

- $\min(i, \max((3-j), 0))$ vektorov z bunky J_2

- $\min(i, \max((2-j), 0))$ vektorov z bunky J_3